Tathagata Debnath

Ph.D. Candidate, Computer Science (NMSU)

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Skills

LLM Fine-Tuning

PyTorch / Hugging Face

Multi-Agent / LangGraph / MCTS

Multi-GPU (H100 / A100 MIG)

RAG / FAISS / Prompt Eng.

Quantization/ vLLM

Classical ML (CNN, RNN, GAN)

Python / R / C++

Education

Ph.D., Computer Science
NMSU GPA 4.00 2018–2026

M.S., CS (AI/ML)
NMSU 2024–2026

M.Tech, CSE (Gold Medal)
Tripura U. 9.83/10 2017

B.Tech, CSE
NIT Agartala 7.62/10 2015

Summary

PhD candidate building LLM systems end-to-end—fine-tuning, agentic orchestration, and RAG. **Agentic AI:** LangGraph multi-agent orchestration, MCTS action search, LLM-as-Judge loops feeding preferences back into DPO. **Prompt engineering:** dual-temperature generation/critique, Pydantic-structured output, chain-of-thought decomposition. Built **CodeQ** (MCTS+DPO, **91% fix rate on DebugBench**) and **BioLite Suite** (biology SLMs). IEEE TPAMI / TCBB author with **81+ citations** and 2 CRAN R packages.

Technical Stack

ML/DL: PyTorch, LoRA/QLoRA, DPO/RLHF/SFT, Hugging Face (Transformers, TRL, PEFT), Flash Attention 2, bitsandbytes (NF4), vLLM, scikit-learn, XGBoost, SHAP
Agents: LangGraph, LangChain, Claude/OpenAI APIs, RAG, FAISS, MCTS **Infra:** Multi-GPU (2×H100, NCCL), MIG A100, Docker, W&B, LangSmith, FastAPI
Languages: Python, R, C/C++, JavaScript, SQL, Bash

Selected Projects

CodeQ: Self-Improving Debugging Agent — 2025

PyTorch, DPO, MCTS, LoRA, 2×H100

- MCTS + DPO on Qwen2.5-Coder-7B; **84% fix rate on DebugBench** (base 43.9%, MCTS-only 81.3%).
- Built Fp32LogitsDPOTrainer to fix bf16 NaN overflow in DPO logits.

BioLite Suite: Domain-Specialized SLMs — 2026 – Present

QLoRA, SFT, DPO, Llama 3.2, MIG A100

- Llama 3.2 (1B/3B) + QLoRA on MIG-partitioned A100 (∼4.8 GB) for biology narratives.
- Released 2 Hugging Face models, 1 dataset, full training code.

Multi-Agent Code Generation (V1) — 2024

Python, Claude API, Pydantic

- Baseline coder/tester multi-agent loop with shared Pydantic state and revision iterations.
- Foundation for the Parallel and Self-Evolving variants.

Parallel Multi-Agent Code Generation — 2024

LangGraph, Claude API, Docker, Streamlit

- Concurrent LangGraph coder workers with Docker-sandboxed pytest (**79/79 pass rate**).
- Streamlit UI with Haiku/Sonnet/Opus selection; LangSmith observability.

Self-Evolving Code Generator — 2024

LangGraph, Claude API, LLM-as-Judge

- LLM-as-Judge scores correctness/efficiency/style and auto-refines prompts.
- Precursor to CodeQ's DPO self-improvement paradigm.

Federated Diabetes Predictor — 2024

Python, Flower, PyTorch, FedAvg

- Federated learning on Diabetes Health Indicators across distributed clients without centralizing data.
- Compared convergence and accuracy vs. centralized baselines.

Selected Publications

- Chen Y., **Debnath T.**, Cai A., Song M. "Circular Silhouette and a Fast Algorithm." *IEEE TPAMI*, 45(11), 2023.
- **Debnath T.**, Song M. "Fast Optimal Circular Clustering on Round Genomes." *IEEE/ACM TCBB*, 18(6), 2021.
- Bendzus-Mendoza H. et al. (incl. **Debnath T.**). "Radiation-induced alternative splicing in *Aedes aegypti*." *Scientific Reports*, 2025.

Awards: M.Tech Gold Medal (Tripura Univ.) · Biopattern Award, NMSU · Ph.D. Tuition Scholarship, NMSU. **CRAN:** OptCirClust, CircularSilhouette.